Certification of Imitation Controllers Using Learned Lyapunov Functions

Master thesis by Josua Christoph Lindemann
Date: June 28, 2026, Date of submission: 16. August 2026

1. Review: Prof. Dr.-Ing. Rolf Findeisen
2. Review: Hendrik Alsmeier M.Sc.
Darmstadt

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3.1 Discrete-Time Dynamical Systems and Stability

To implement an optimal control scheme on digital hardware, the underlying continuous-time system dynamics

$$\dot{x}(t) = f(x(t), u(t)) \tag{3.1}$$

must be discretized, where $x(t) \in \mathcal{X}$ and $u(t) \in \mathcal{U}$ denote the continuous-time state and input vectors, respectively. In this work, the corresponding discrete-time system

$$x_{k+1} = f_d(x_k, u_k) \tag{3.2}$$

is obtained by approximating the continuous dynamics using a numerical integration scheme with a constant sampling time T_s, such as the forward Euler or a fourth-order Runge-Kutta (*RK4*) method **something**.

3.1.1 Lyapunov Stability

Let the origin $(x, u) = (\mathbf{0}, \mathbf{0})$ be an equilibrium point of the system such that $f_d(\mathbf{0}, \mathbf{0}) = \mathbf{0}$. Suppose the system is controlled by a state-feedback control policy $u_k = \kappa(x_k)$. The closed-loop system is said to be *Lyapunov stable* if there exists a positive definite Lyapunov function $V(x_k)$ such that:

$$\begin{aligned} V(\mathbf{0}) &= 0 & & \\ V(x_k) &> 0 & \quad & \forall x_k \in \mathcal{X} \setminus \{\mathbf{0}\} \\ V(f_d(x_k, \kappa(x_k))) - V(x_k) &\leq 0 & \quad & \forall x_k \in \mathcal{X} \setminus \{\mathbf{0}\}. \end{aligned} \tag{3.3}$$

Furthermore, the equilibrium point is *locally asymptotically stable* if the Lyapunov function additionally satisfies a strictly negative definite decrease condition along the closed-loop trajectories:

$$V(f_d(x_k, \kappa(x_k))) - V(x_k) < 0 \qquad \forall x_k \in \mathcal{X} \setminus \{\mathbf{0}\}. \tag{3.4}$$

Asymptotic stability is typically the desired property for control systems, as it ensures that the system will converge to the equilibrium point over time.

3.2 Model Predictive Control and Stability

3.2.1 Problem Formulation

$$
\begin{aligned}
\{x_k^*\}_0^N, \{u_k^*\}_0^{N-1} = \underset{\{x_k\}_0^N, \{u_k\}_0^{N-1}}{\operatorname{argmin}} & \sum_{k=0}^{N-1} \ell(x_k, u_k) + V(x_N) \\
\text{s.t.} \quad & x_{k+1} = f(x_k, u_k) && \forall k \in [0, \dots, N-1] \\
& x_k \in \boldsymbol{\mathcal{X}_k} && \forall k \in [0, \dots, N] \\
& u_k \in \boldsymbol{\mathcal{U}_k} && \forall k \in [0, \dots, N-1] \\
& x_0 = x_{init}.
\end{aligned} \tag{3.5}
$$

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7.1 System Modeling and Problem Formulation

For our experimental setup, two benchmark systems are selected: the double integrator and the cartpole. These systems provide a good balance between simplicity and complexity allowing us to certify the stability of the closed-loop system.

7.1.1 Double Integrator

The double integrator is a simple second order linear system, therefore it serves as the ideal baseline system in the learning and certification pipeline. The continuous-time dynamics are described by the following state-space equation:

TODO: FIX to Discrete-Time Dynamics

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t) \tag{7.1}$$

where $x(t) = \begin{bmatrix} p(t) & v(t) \end{bmatrix}^\top$ is the state vector containing the position and velocity of the system, respectively, and $u(t)$ denotes the scalar control input (acceleration).

The *MPC setup* is formulated according to the problem in Eq. (3.5) subject to the double-integrator dynamics in Eq. (7.1). A prediction horizon of $N = 20$ steps is chosen, and the states and inputs are penalized using the weighting matrices $Q = \text{diag}(15, 1)$ and $R = 0.1$.

7.1.2 Cartpole

7.2 Linear System: Double Integrator

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